HAOLIN ZOU

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Research Profile

Research: I develop statistical theory and efficient algorithms for machine learning in high-dimensional settings, with a focus on **machine unlearning**, **risk estimation**, and **data attribution**. I study how statistical models can be efficiently updated, evaluated, and interpreted, and establish theoretical guarantees for these procedures.
Additional Interests: network models, decoupling and self-normalization, benign overfitting, AI memory.

Academic and Research Appointments

- **Chinese University of Hong Kong** *July 2026 - Present*
  Postdoctoral Researcher. Advisor: Prof. Junhui Wang.
  - Developing statistical theory and computational methods for machine unlearning and risk estimation in high-dimensional models.
  - Investigating computationally efficient dyad- and node-level data attribution in statistical network models.
- **Voleon Capital Management** *May 2025 - Aug 2025*
  Quantitative Research Intern
  - Developed and evaluated quantitative methods for bond pricing and model comparison.

Education

- **Columbia University** *Sep 2020 - May 2026*
  Ph.D. in Statistics. Advisor: Prof. Arian Maleki, Prof. Victor de la Peña. New York, USA
- **Columbia University** *Sep 2018 - Dec 2019*
  M.S. in Actuarial Science New York, USA
- **Peking University** *Sep 2014 - Jul 2018*
  Bachelor of Applied Mathematics, Bachelor of Economics (double major) Beijing, China

Publications

- **Journal Articles**
  - **Zou, H.**, Yao, H., and de la Peña, V. H. (2026). Exact Moment Formulae for Mean-normalized Statistics with Applications to Inequality Measures. *Statistics and Probability Letters*, 236:110754.
  - **Zou, H.**, Auddy, A., Rahnama Rad, K., and Maleki, A. (2025). Certified Machine Unlearning Under High Dimensional Regime. *Journal of Machine Learning Research (JMLR)*, 26(308), 1-58.
  - Baydil, B., de la Peña, V., **Zou, H.**, and Yao, H. (2025). Unbiased Estimation of The Gini Coefficient. *Statistics and Probability Letters*, 222:110376.
  - Auddy, A., **Zou, H.**, Rahnama Rad, K., and Maleki, A. (2024). Approximate Leave-one-out Cross Validation for Regression with L1 Regularizers. *IEEE Transactions on Information Theory*, 70(11):8040-8071.
  - de la Peña, V., Gzyl, H., Mayoral, S., **Zou, H.**, and Alemayehu, D. (2024). Prediction And Estimation of Random Variables with Infinite Mean Or Variance. *Communications in Statistics - Theory and Methods*, 54(1):1-15.
- **Conference Proceedings**
  - Tong, H., Ghosh, S., **Zou, H.**, and Maleki, A. (2026). Imperfect Influence, Preserved Rankings: A Theory of TRAK for Data Attribution. *Accepted at NeurIPS 2026*.
  - Pandey, A., Auddy, A., **Zou, H.**, Maleki, A., and Kulkarni, S. (2026). Gaussian Certified Unlearning in High Dimensions: A Hypothesis Testing Approach. *ICLR 2026*. **[Oral; top 4% of accepted papers.]**
  - **Zou, H.**, Auddy, A., Rahnama Rad, K., and Maleki, A. (2025). Theoretical Analysis of Leave-one-out Cross Validation for Non-differentiable Penalties under High-dimensional Settings. *AISTATS 2025*, PMLR 258:4033-4041.
  - Auddy, A., **Zou, H.**, Rahnama Rad, K., and Maleki, A. (2024). Approximate Leave-one-out Cross Validation for Regression with L1 Regularizers. *AISTATS 2024*, PMLR 238:2377-2385. **[Oral; top 7% of accepted papers.]**
- **Workshop Papers**
  - **Zou, H.**, Auddy, A., Kwon, Y., Rahnama Rad, K., and Maleki, A. (2025). Newfluence: Boosting Model Interpretability and Understanding in High Dimensions. *ICML 2025, Workshop: Assessing World Models: Methods and Metrics for Evaluating Understanding.*
- **Manuscripts and Ongoing Research**
  - **Zou, H.** and Wang, J. (2026). Dyad and Node Attribution for Statistical Network Models. *Ongoing Project.*
  - Shang, C., **Zou, H.**, Zhu, B., and Maleki, A. (2026). Error Accumulation in Second Order Sequential Machine Unlearning. *Submitted to AISTATS 2027.*
  - **Zou, H.**, Rahnama Rad, K. and Maleki, A. (2026). The Bias and Variance of K-fold Cross Validation in High Dimensions. *Manuscript in preparation.*

Talks and Presentations

- **Invited Seminar Talks**
  - City University of Hong Kong. *Certified Machine Unlearning in High Dimensions* (Jan 2026).
  - Chinese University of Hong Kong, Shenzhen. *Machine Unlearning under High Dimensional Regime* (Jan 2026).
  - Columbia University. *High-dimensional Asymptotic Analysis: Opportunities and Challenges* (Nov 2024).
- **Conference and Workshop Presentations**
  - Columbia University, Minghui Yu Memorial Conference. *Approximate Certified Data Removal in High Dimensional R-ERM* (Apr 2025).
  - AISTATS 2024, oral presentation. *Approximate Leave-one-out CV for Regression with L1 regularizers* (May 2024).
  - Columbia University, Minghui Yu Memorial Conference. *Approximate LOOCV in High Dimensions* (Apr 2024).
  - INFORMS 2022, oral presentation. *The Bias of Gini Coefficient* (Oct 2022).

Teaching Experience

- **Recitation Instructor** *Jan 2024 - May 2024*
  Columbia University
  - Led weekly recitations for Statistical Inference and Modeling, developing review materials and problem-solving sessions for graduate students.
- **Co-instructor** *Apr 2024*
  Georgia Institute of Technology
  - Co-instructed with Prof. Victor de la Peña on the applications of decoupling and self-normalized inequalities to bandit and sorting problems in a short course.
- **Teaching Associate** *Sep 2020 - May 2026*
  Columbia University
  - Core MS course: Probability Theory (2 semesters), Statistical Inference (4 semesters)
  - Advanced methods: Generalized Linear Models (1 semester)
  - Undergraduate course: Introduction to Statistics (4 semesters)

Professional Service

- **Journal Reviewer**
  - Annals of Applied Probability
  - Journal of the Royal Statistical Society, Series B
  - IEEE Transactions on Information Theory
- **Conference Reviewer**
  - NeurIPS, ICML, AISTATS.
- **Session Chair**
  - INFORMS, Session "Heavy-tailedness, Dependence and Robustness" (Oct 2022).

Awards and Distinctions

- **Second Prize** *Sep 2017*
  China Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM)
- **First Class Scholarship** *Sep 2017*
  Yizheng Alumni Scholarship
- **First Prize** *Oct 2014*
  China Undergraduate Contest in Physics

Additional Information

Coding and Computing: Python (advanced), R, Stata, Stan.
Languages: English (fluent), Chinese (native), Japanese (proficient).